

Simplified stress analysis of conventional fibrous composite laminates subjected to in-plane loads

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April 11, 2020

Abstract

A simplified methodology for calculating the strain level in a fibrous composite laminate subjected to in-plane loads, provided it complies with conventional design rules and is built from the legacy base of angles 0° , $\pm 45^\circ$ and 90° , is presented, and used, combined with the *Maximum Truncated Strain* criterion, to predict approximately the strength of the laminate.

1 Introduction

Predicting the ultimate strength of fibrous composite laminates is one of the most significant problems for working efficiently with this kind of materials. A problem which has not been definitively solved indeed, even the combination of Finite Elements Analysis and progressive failure criteria is showing an enormous potential, being one of the most popular topic of research on structural materials along past years.

However, yet nowadays, the extensive use of sophisticated numerical methodologies can be still unpractical in many circumstances for the daily work of a stress engineer, in particular in the aerospace industry. Under these circumstances (pre-sizing, optimization, tests preparation...) the fast and explicit solutions offered by classical handbooks and literature, combined with the internal loads analysis performed with global Finite Elements Models, and even with the handicap of that limited accuracy, prevail as the battle horse along many phases of the design process of airframes.

The purpose of this paper is to present a simplified methodology to calculate the strain in a laminate (and its conforming plies) under static in-plane loads, when laminate is built using the traditional base of angles (0° , $\pm 45^\circ$

and 90°) following good design practices (balanced, symmetric, ...), in order to predict, approximately, its strength in a later step. It is admitted that this method is merely an adaptation, to this specific work scenario, of available *Classical Laminate Theory* and pre-existing failure theories, and not really an original work. Its merit, if any, is precisely the explicit simplifications done.

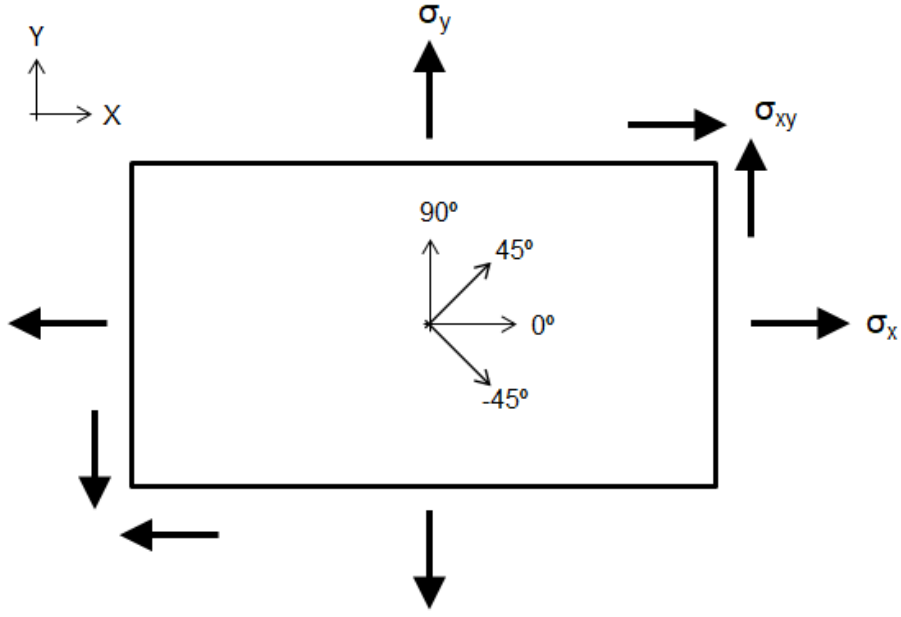


Figure 1: Reference axis system for laminate, plies and loads

2 Analysis method

As noted above, the laminate considered shall follow good design practices (balanced, symmetric, ...) and be constructed using the legacy base of angles 0° , $\pm 45^\circ$ and 90° for its stacking sequence. Without loss of generality (for the in-plane/membrane analysis), a laminate of this kind can be defined by its thickness and the proportions of plies at 0° , $\pm 45^\circ$ and 90° directions.

$$w_0 = n_0/n$$

$$w_{45} = n_{45}/n$$

$$w_{90} = n_{90}/n = 1 - w_0 - w_{45}$$

2.1 Stiffness and elastic moduli of the laminate

In a first step of the analysis, the stiffness of the laminate shall be calculated for later stating the basic equation of the *Classical Laminate Theory*. The reduced stiffness matrix of the orthotropic ply (or the quasi-orthotropic laminate) has the shape:

$$[Q] = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}$$

From Jones' page 72 (Ref. [1]), the relationships between reduced stiffness of the lamina (reference is 0° material, or X-laminate, direction) and the material engineering constants (elastic moduli) can be expressed as:

$$\begin{aligned} Q_{11}^0 &= E_1 / (1 - \nu_{12} \cdot \nu_{21}) & \nu_{12} / E_1 &= \nu_{21} / E_2 \\ Q_{22}^0 &= E_2 / (1 - \nu_{12} \cdot \nu_{21}) \\ Q_{12}^0 &= \nu_{12} E_2 / (1 - \nu_{12} \cdot \nu_{21}) = \nu_{21} E_1 / (1 - \nu_{12} \cdot \nu_{21}) \\ Q_{66}^0 &= G_{12} \end{aligned}$$

From the same reference, Jones' page 77, it is possible to get the reduced stiffness, referred to the global laminate axis system, also for the plies-pairs at $+45^\circ/-45^\circ$ as:

$$\begin{aligned} Q_{11}^{45} &= Q_{22}^{45} = 0.25 \cdot [Q_{11}^0 + 2 \cdot (Q_{12}^0 + 2 \cdot Q_{66}^0) + Q_{22}^0] \\ Q_{12}^{45} &= 0.25 \cdot (Q_{11}^0 + Q_{22}^0 - 4 \cdot Q_{66}^0) + 0.5 \cdot Q_{12}^0 \\ Q_{66}^{45} &= 0.25 \cdot (Q_{11}^0 + Q_{22}^0 - 2 \cdot Q_{12}^0 - 2 \cdot Q_{66}^0) + 0.5 \cdot Q_{66}^0 \\ Q_{16}^{45} &= Q_{26}^{45} = 0 \quad (\text{provided } +45^\circ \text{ and } -45^\circ \text{ plies come in pairs}) \end{aligned}$$

Plies at 90° shall have the same stiffness matrix as 0° plies but interchanging Q_{11} and Q_{22} terms.

Then the reduced stiffness matrix for the laminate can be obtained summing the previous matrices weighted by the respective proportions of the plies at 0° , $\pm 45^\circ$ and 90° directions within the laminate:

$$[Q] = w_0 \cdot [Q]^0 + w_{45} \cdot [Q]^{45} + w_{90} \cdot [Q]^{90}$$

Note that the apparent elastic moduli of the laminate can be derived from the resulting $[Q]$ matrix (equal to $[A]/t$) using the expressions presented above for the ply, but expressed as $E_i = f(Q_{ij})$.

$$\begin{aligned}\nu_{xy} &= Q_{12}/Q_{22} \\ E_x &= (Q_{12})^2/Q_{22} \\ E_y &= (Q_{12})^2/Q_{11} \\ G_{xy} &= Q_{66}\end{aligned}$$

2.2 Strain in laminate and its plies

The mid-plane strain is obtained using the Classical Lamination Theory from the applied loads (in-plane forces per unit width) modifying slightly the familiar expression $[N] = [A][\varepsilon]$, for dealing with apparent stresses instead of plate forces, as:

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_{xy} \end{bmatrix} = [Q] \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} \Rightarrow \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} = [Q]^{-1} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_{xy} \end{bmatrix}$$

Strains ε_x , ε_y , γ_{xy} can be obtained from expression above, where ε_x and ε_y are equal to the strain at 0° and 90° plies respectively. The strain levels at 45° plies can be obtained also from values of ε_x , ε_y , γ_{xy} and the appropriate transformation equations (see Jones' 2.73). Or, expressed via the engineering constants, without the matrix notation:

$$\begin{aligned}\varepsilon_0 &= \varepsilon_x = \sigma_x/E_x - \nu_{xy} \cdot \sigma_y/E_x \\ \varepsilon_{90} &= \varepsilon_y = \sigma_y/E_y - \nu_{xy} \cdot \sigma_x/E_x \\ \gamma_{xy} &= \sigma_{xy}/G_{xy} \\ \varepsilon_{45} &= (\varepsilon_x + \varepsilon_y + \gamma_{xy})/2 \\ \varepsilon_{-45} &= (\varepsilon_x + \varepsilon_y - \gamma_{xy})/2\end{aligned}$$

2.3 Strength of the laminate

The selected failure criterion for the laminate is the *Maximum Truncated Strain* criterion proposed by Hart-Smith and suggested by the CMH-17 Vol. 3 Chpt. 8 (Ref. [2]). Its simplicity is consistent with the motivation of this method. Nevertheless, it is strongly recommended that the allowable strain values used for the analysis do not rely exclusively on the tests results for the uni-directional laminate.

For plies at $0^\circ/90^\circ$ and for plies at $+45^\circ/-45^\circ$ it must be checked that:

$$\begin{aligned}\varepsilon_{1c} &< \varepsilon_1 < \varepsilon_{1t} \\ \varepsilon_{1c} &< \varepsilon_2 < \varepsilon_{1t} \\ |\varepsilon_1 - \varepsilon_2| &< (1 + \nu_{12}) \cdot \max(\varepsilon_{1t}, \varepsilon_{1c})\end{aligned}$$

Material allowable values to be use in expressions above:

- Tension allowable strain ε_{1t} shall be preferably corrected with results available from plain strength tests in multi-angular laminates, e.g. in the quasi-isotropic $(45/-45/90/0)_N S^1$
- Compression allowable strain ε_{1c} can be directly obtained from the longitudinal compression test in the UD laminate as F_{1c}/E_1
- For bi-axial fabric materials ($E_1 = E_2$), instead of the Poisson's ratio of the material γ_{12} , use directly a value of 0.3 (i.e. treat the fabric ply as a cross-ply $0/90$ sublaminates made of a fictitious UD tape material with the same allowable strains)

3 Conclusions

This paper presented a simple and fast method for checking the plain strength of composite laminates (following the traditional design rules) under static in-plane loads.

An implementation of this methodology is available in a Microsoft Excel spreadsheet named *ANCOSheet*. The distribution and use of this software is permitted under the terms of the BSD License 2.0. The following

¹Using the allowable tension strain obtained from the tension test on the QI laminate might NOT be conservative for laminates with less than 25% plies in tension load direction

figures exemplify the use of this spreadsheet for calculating the laminate elastic moduli and for predicting its plain strength. Material data in this example are taken from the NIAR qualification tests report for the Hexcel AS4/8552 UD pre-preg CFRP material (Ref. [3]): mean values at RTD conditions, converted to the SI units (MPa), Young's modulus (see Figure 2) as the average of the values from tension and compression results in UD tests (rounded to 2-3 significant digits and, in longitudinal direction, normalized with the nominal ply thickness), allowable strengths according to recommendations in previous section (ε_{1t} from the QI laminate tension strength, ε_{1c} from the longitudinal compression strength). Notice the presented static margins (the Reserve Factors indeed, RF values in Figure 3) correspond to the unnotched plain strength of the (50/40/10) *hard* laminate and show reasonably conservative predictions.

ELASTIC MODULI OF THE LAMINATE FROM LAMINA PROPERTIES AND PLIES WEIGHTS AT 0°, ±45°, 90° DIRECTIONS																			
Elastic moduli of the lamina																			
E_1	E_2	G_{12}	ν_{12}	ν_{21}	$s=c=\sin(45)$ $=\cos(45)$	$s^4=c^4$ $=s^2 \cdot c^2$													
119000	9500	4800	0.32	0.02555	0.7071	0.25													
Stiffness matrix of the lamina at directions 0°, 90° and at 45° (last one at +45/-45 indeed)																			
$Q_0 =$	<table><tr><td>119981</td><td>3065</td><td>0</td></tr><tr><td>3065</td><td>9578</td><td>0</td></tr><tr><td>0</td><td>0</td><td>4800</td></tr></table>			119981	3065	0	3065	9578	0	0	0	4800							
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$Q_{45} =$	<table><tr><td>38722</td><td>29122</td><td>0</td></tr><tr><td>29122</td><td>38722</td><td>0</td></tr><tr><td>0</td><td>0</td><td>30857</td></tr></table>			38722	29122	0	29122	38722	0	0	0	30857							
38722	29122	0																	
29122	38722	0																	
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Elastic moduli of the laminate defined by plies relative weights at 0°, ±45°, 90° directions																			
Style	W_{0°	W_{45°	W_{90°	Q_{11}	Q_{22}	Q_{12}	Q_{66}	E_x	E_y	G_{xy}	ν_{xy}								
Soft	10%	80%	10%	43934	43934	23911	25646	30920	30920	25646	0.5442								
QI	25%	50%	25%	51751	51751	16094	17829	46746	46746	17829	0.3110								
Hard1	40%	40%	20%	65397	43316	13488	15223	61197	40535	15223	0.3114								
Hard2	50%	40%	10%	76437	32276	13488	15223	70801	29896	15223	0.4179								
Hard3	60%	30%	10%	84563	29362	10882	12617	80530	27961	12617	0.3706								

Figure 2: Apparent elastic moduli of Hexcel AS4/8552 calculated with *ANCOsheet*

Note that the laminate/plate failure due to loss of stability (buckling) under compression or shear loads is out of the scope of this method but shall have to be checked anyway.

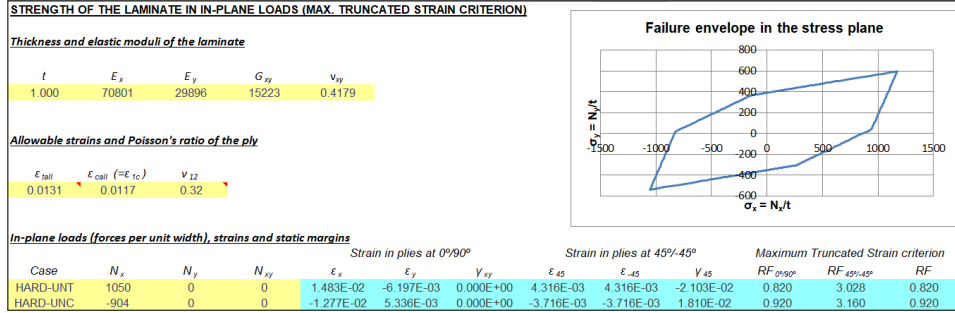


Figure 3: Plain strength of a *hard* (50/40/10) laminate made of Hexcel AS4/8552 predicted by *ANCOsheet*

Nomenclature

- 1 As superscript, local longitudinal direction of the ply
 - 2 As superscript, local transverse direction of the ply
- $[A]$ In-plane stiffness matrix of the laminate
- $[Q]$ Reduced stiffness matrix of the laminate/ply, $[Q] = (1/t) \cdot [A]$
- γ Engineering shear strain
- ν_{12} Poisson's ratio of the ply material
- σ Stress (note the apparent stress in the laminate is $[\sigma] = (1/t) \cdot [N]$)
- ϵ (Axial) strain
- ϵ_{1c} Allowable compression strain
- ϵ_{1t} Allowable tension strain
- CFRP* Carbon Fibre Reinforced Polymer
- E_1 Elastic modulus of the ply material in longitudinal direction
- E_2 Elastic modulus of the ply material in transverse direction
- FEM* Finite Elements Model
- G_{12} In-plane shear modulus of the ply material
- N Internal loads in laminate (force per unit width)
- n Total number of plies in the laminate
- n_0 Number of plies oriented at 0° direction

n_{45}	Number of plies oriented at $\pm 45^\circ$ directions
n_{90}	Number of plies oriented at 90° direction
QI	Quasi-Isotropic, referred to a kind of laminates, made typically of packages of (45/-45/90/0) sub-laminates
RF	Reserve Factor, i.e. ration between allowable load and applied load
SI	International Units System
t	Thickness of the laminate
UD	Uni-Directional (material or laminate)
w_0	Proportion of plies oriented at 0° direction
w_{45}	Proportion of plies oriented at $\pm 45^\circ$ directions
w_{90}	Proportion of plies oriented at 90° direction
x	As supscript, global longitudinal direction of the laminate, typically matching material reference direction 0°
y	As supscript, global transverse direction of the laminate

References

- [1] MECHANICS OF COMPOSITE MATERIALS, 2nd Edition, by R. M. Jones
- [2] COMPOSITE MATERIALS HANDBOOK (CMH-17), Edition 2012, by the FAA/SAE
- [3] Hexcel 8552 AS4 Unidirectional Prepreg at 190 gsm & 35% RC Qualification Material Property Data Report (NCAMP Test Report Number: CAM-RP-2010-002 Rev A), May 2011, by the NIAR